



Smart Material–Driven Adaptive Drilling Mechanism Using Helical Nitinol Springs for Lunar Surface Drilling

MECHANISM & COMPONENTS

Helical Nitinol (NiTi) SMA Springs



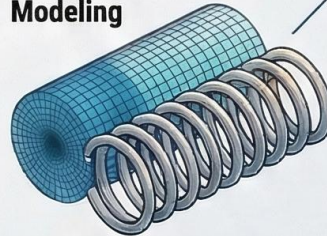
Martensite



Austenite

These components provide passive axial thrust and enable radial self-anchoring within the drill hole.

Lagoudas Constitutive Modeling



Thermomechanical behavior modeled to ensure stress and strain remain within operational limits during repeated cycling.

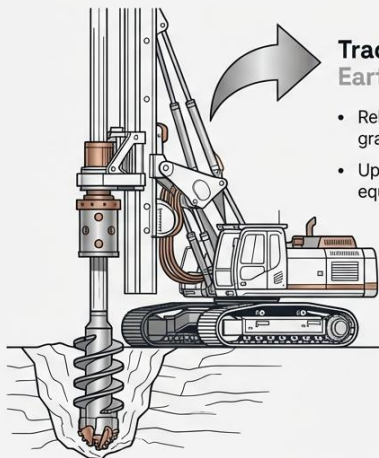


Team Members

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Scope and Objective



Traditional Earth Drilling

- Relies on gravity/weight-on-bit
- Up to 60 kN of equivalent axial force

Why? A fundamentally different mechanism is required to generate internal thrust and anchoring force without relying on heavy structural mass or fluid hydraulics for **LUNAR DRILLING**.

Objective: Develop a mechanism to generate 700–900 N internal thrust and passive anchoring forces



Lunar Environment Constraints

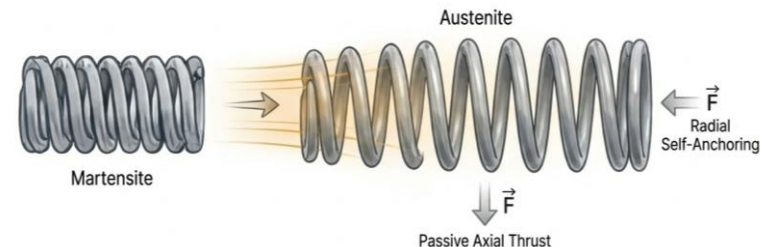
- High vacuum & severe mass constraints
- Fraction of Earth's gravitational acceleration



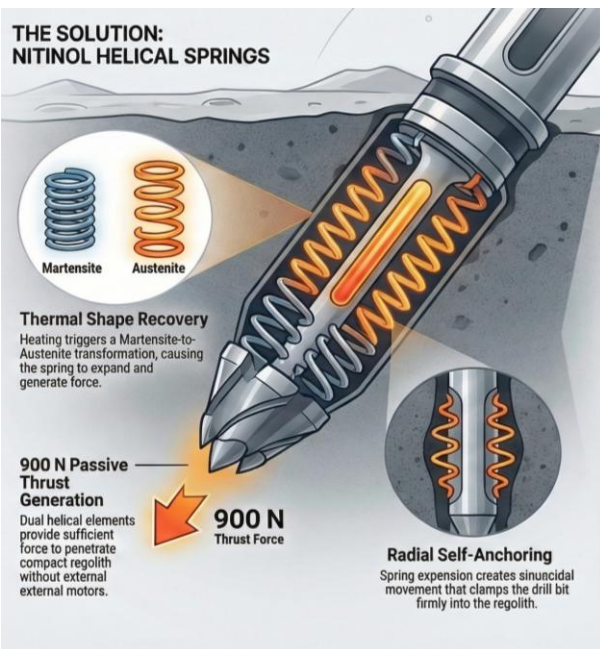


Methodology

Core Model: Lagoudas constitutive modeling used to simulate reversible solid-state phase transformations (Martensite to Austenite)



THE SOLUTION: NITINOL HELICAL SPRINGS



SMA-Driven Adaptive Drilling

The proposed solution integrates helical Nitinol (NiTi) shape memory alloy springs directly along the drill shaft. This generates substantial internal thrust and anchoring forces without the mass or complexity of motors, hydraulics, or pneumatic actuation systems.





Approach

Optimized cutting geometry built for extreme material resistance

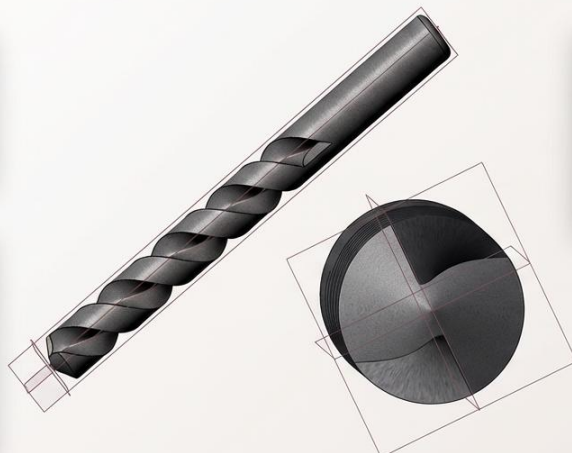
The system couples the Nitinol thrust mechanism with a specialized cutting interface designed for low-power, high-efficiency penetration.

Base Material:

Titanium alloy (Ti-6Al-V) with tool steel structural components.

Coating:

Chromium Carbide (Cr₃C₂-NiCr) and Titanium Carbide (TiC) reinforced cutting tips to withstand regolith abrasion.



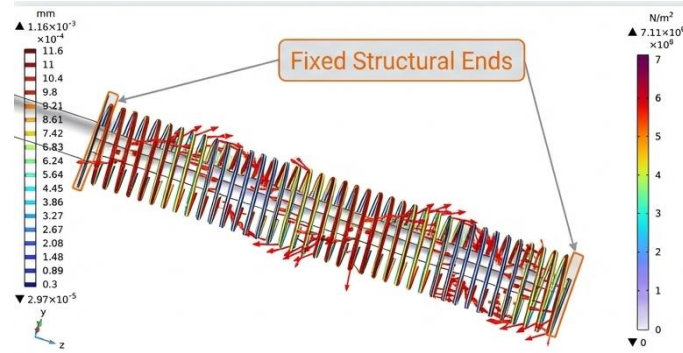
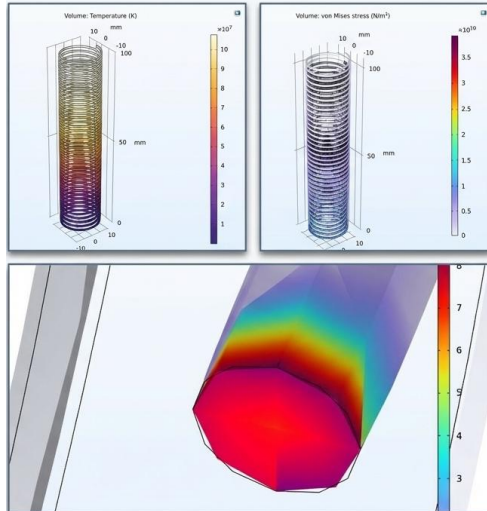
Geometry:

A double flute bit engineered with a highly specific 2.28-degree flute angle to clear fine particles in a vacuum.



Approach

- The system integrates helical Nitinol (NiTi) shape memory alloy (SMA) mechanisms along the drill shaft to generate mechanical work through reversible martensitic–austenitic phase transformations.
- Material thermo-mechanical behavior is simulated utilizing the experimentally validated Lagoudas constitutive model.



$$0 = \nabla \cdot \mathbf{S} + \mathbf{F}_V$$

$$\mathbf{S} = \mathbf{S}_{inel} + \mathbf{S}_{el}, \quad \epsilon_{el} = \epsilon - \epsilon_{inel}$$

$$\mathbf{S}_{el} = \mathbf{C} : \epsilon_{el}$$

$$\mathbf{S}_{inel} = \mathbf{S}_0 + \mathbf{S}_{ext} + \mathbf{S}_q$$

$$\epsilon_{inel} = \epsilon_0 + \epsilon_{tr} + \epsilon_{ext} + \epsilon_{th}$$

$$\epsilon = \frac{1}{2} [(\nabla \mathbf{u})^T + \nabla \mathbf{u}]$$

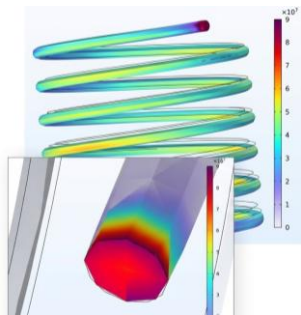
$$\dot{\epsilon}_{tr} = \dot{\xi} \Lambda, \quad \dot{\xi} = \pm \lambda, \quad \lambda \geq 0, \quad F(\mathbf{S}, \sigma_{ys}) \leq 0, \quad \lambda F = 0$$

$$F = \begin{cases} \Pi - \sigma_{ys}, & \dot{\xi} > 0 \\ -\Pi - \sigma_{ys}, & \dot{\xi} < 0 \end{cases}, \quad \Lambda = \begin{cases} \frac{3}{2} \epsilon_{tr, \max} \frac{\text{dev}(\mathbf{S})}{\|\text{dev}(\mathbf{S})\|}, & \dot{\xi} > 0 \\ \epsilon_{tr, \max} \frac{\epsilon_{tr-r}}{\|\epsilon_{tr-r}\|}, & \dot{\xi} < 0 \end{cases}$$

$$\Pi(\mathbf{S}, T, \xi) = s : \Lambda + \frac{1}{2} s : \Delta \mathbf{s} : s + s : \Delta \alpha (T - T_{ref}) - \rho \Delta c \left[(T - T_0) - T \ln \left(\frac{T}{T_0} \right) \right] + \rho \Delta s_0 T - \frac{\partial f}{\partial \xi}$$



Conclusion



700–900 N

Total combined thrust force

350–450 N

Axial force per helical spring

Equilibrium heating of the dual helical SMA springs produces peak force transfer values more than sufficient to penetrate compacted lunar regolith without any external loading.

PERFORMANCE & SIMULATION RESULTS



700–900N

Generated Thrust

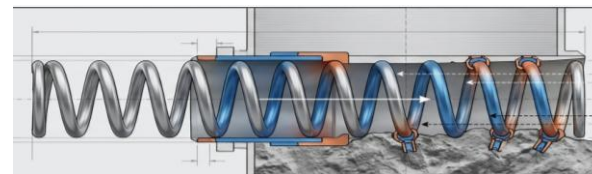
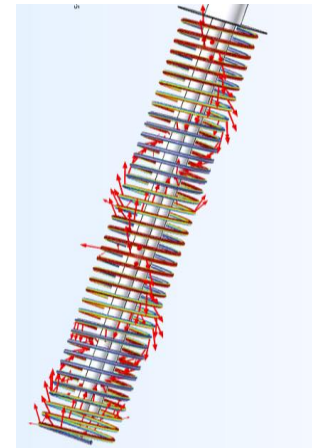
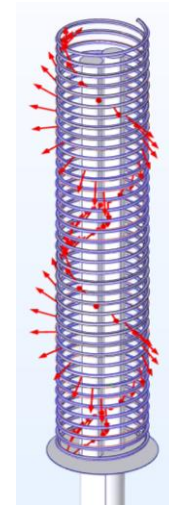
Simultaneous operation of two helical SMA springs provides sufficient forces for regolith penetration.



Integrated Self-Anchoring

Controlled sinusoidal expansion of the springs clamps the drill bit into the soil.

Passive Self-Anchoring→Nitinol expansion acts as a regolith anchor, ensuring stable drill advancement without external heavy structures.



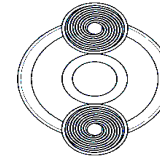


Future Works

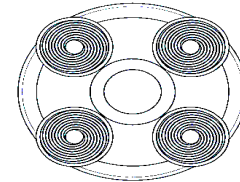
1400–1800 N

Next-generation theoretical thrust capacity

The modularity of the helical bundle design allows for direct linear scaling. By expanding the architecture to incorporate 3 or 4 synchronized Nitinol springs, the system mathematically projects a massive increase in thrust, putting passive lunar drilling on par with heavy Earth operations.



Current Model
(900N)



Scaled Model
(1800N)

Single Element Output:

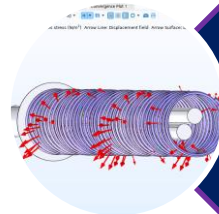
7 MPa → 351 N

9 MPa → 452 N

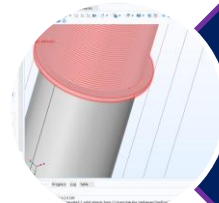


1400–1800 N

System Total Output (4-Spring Bundle)



Thrust Scaling: Increasing capacity to 1400–1800 N using 4-spring bundles to achieve deeper penetration into compact lunar regolith



Fabrication Optimization: Refining Friction Stir Welding for the SMA-to-bit union to consistently maintain 79–87% joint efficiency with refined grain sizes



Thank You!